

REPORT

PCAD LAB

MASTER OF TECHNOLOGY
(POWER ENGINEERING)



Submitted by

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Contents

1	Objective	2
2	Theory	2
3	Three-Phase Inverter Circuit for 50 W BLDC Motor Drive	2
4	Circuit Components	3
5	Angle Generation	3
6	Switching Logic	3
6.1	Gate Signals	3
6.2	Special Case (Wrap-Around Logic)	3
6.3	Logic-1	4
6.3.1	Result-1	5
6.4	Logic-2	6
6.4.1	Result-2	7
6.5	Logic-3	7
7	Simulink Implementation	8
8	Hardware Interface	8
9	Results	8
10	Project Demonstration and Response	9
11	Conclusion	10

1. Objective

- The main objective of this work is to develop and analyze control logic for 120-degree conduction mode operation of a BLDC motor drive system. The work also aims to achieve proper motor operation by implementing an effective switching and control strategy through simulation.

2. Theory

In a 3-phase inverter operating in 120° conduction mode:

- Each switch conducts for 120° electrical
- At any instant, only two switches are ON
- One phase remains OFF (floating)

3. Three-Phase Inverter Circuit for 50 W BLDC Motor Drive

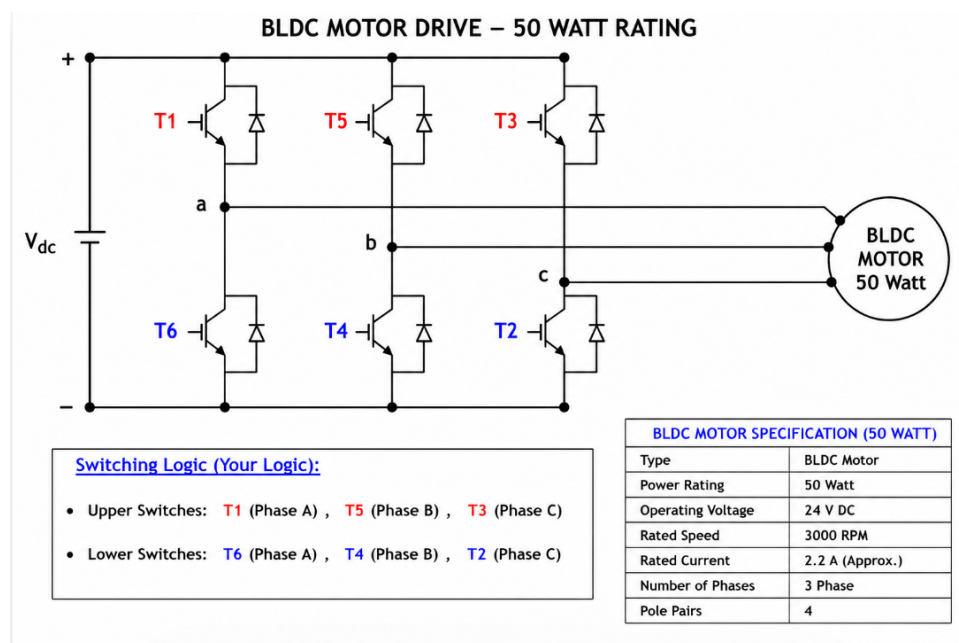


Figure 1: Power Circuit

4. Circuit Components

S. No.	Component	Specification	Quantity
1	Transformer	220 V / 15-0-15 V	4
2	Bridge Rectifier	Full-wave bridge	4
3	Electrolytic Capacitor	470 μ F	4
4	Voltage Regulator IC	LM7815	4
5	PCB	General purpose	2
6	TLP250	Isolation	6
7	Two Way terminal	General purpose	8
8	MOSFET	IRF9530N	6
9	DSP board	F28379D	1

5. Angle Generation

The electrical angle is generated and limited within 0° to 360° using modulo operation:

$$\theta = \text{mod}(\theta, 360) \quad (1)$$

6. Switching Logic

6.1 Gate Signals

$$T1 = (\theta \geq 0) \wedge (\theta < 120) \quad (2)$$

$$T2 = (\theta \geq 60) \wedge (\theta < 180) \quad (3)$$

$$T3 = (\theta \geq 240) \wedge (\theta < 360) \quad (4)$$

$$T5 = (\theta \geq 120) \wedge (\theta < 240) \quad (5)$$

$$T6 = (\theta \geq 180) \wedge (\theta < 300) \quad (6)$$

$$(7)$$

6.2 Special Case (Wrap-Around Logic)

For switch T4, since the conduction interval crosses 360° :

$$T4 = (\theta \geq 300) \vee (\theta < 60) \quad (8)$$

6.3 Logic-1

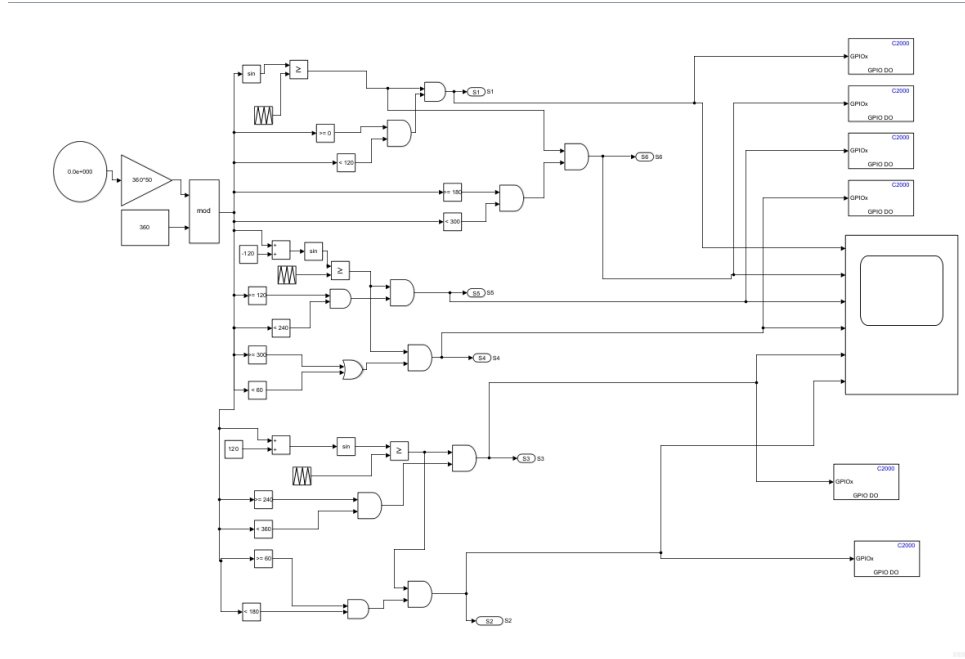


Figure 2: Simulation Circuit

- Initially, the control logic for 120-degree mode operation was developed using Sinusoidal PWM (SPWM) technique. The implementation produced the required output results successfully during simulation. However, after further discussion, Surya Sir suggested that SPWM is not necessary for 120-degree mode operation, because this mode is generally based on six-step commutation switching logic. Therefore, the developed control approach needs to be modified according to the standard 120-degree conduction method.

6.3.1 Result-1

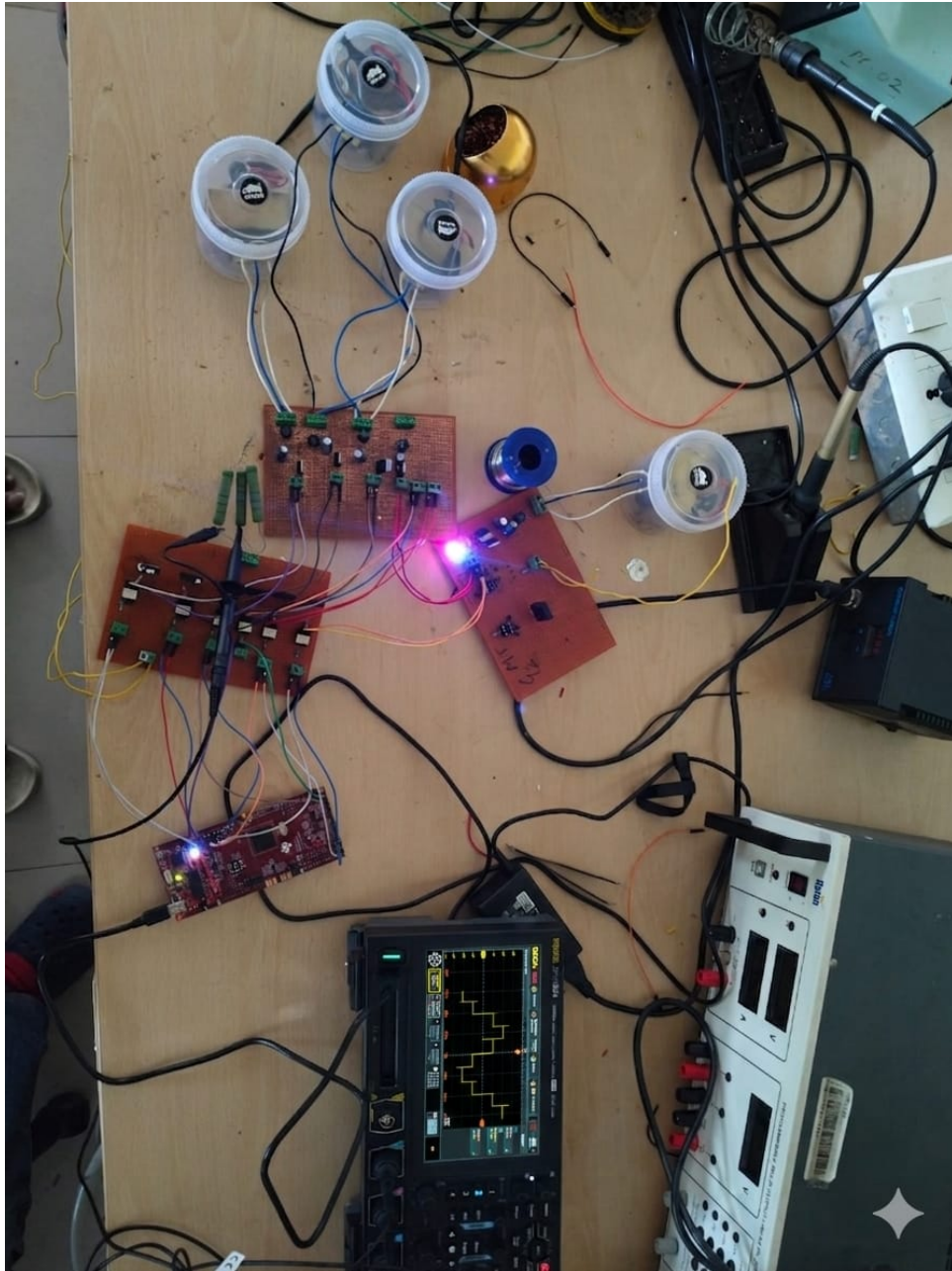


Figure 3: Hardware Circuit with result

6.4 Logic-2

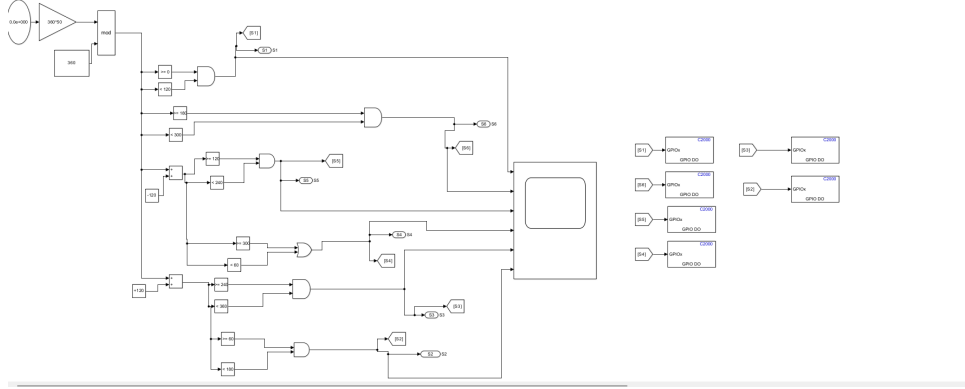


Figure 4: Simulation Circuit

- A second control logic was also developed and tested for the BLDC motor drive in order to improve the operation of the 120-degree conduction mode. Switching sequences and gate pulse generation were successfully implemented according to the required six-step commutation pattern. The simulation results confirmed that the inverter switching logic was functioning properly. However, despite the successful implementation of the commutation strategy, the BLDC motor did not achieve the expected operation and stable speed response. The main reason for this issue was the absence of a proper V/f (Voltage-to-Frequency) control technique in the developed system. Since the applied voltage was not regulated in proportion to the operating frequency, the motor could not maintain the flux and torque characteristics required for smooth operation. Therefore, although the control logic and switching implementation were completed successfully, the overall drive system remained incomplete without the integration of an appropriate V/f control method.

6.4.1 Result-2

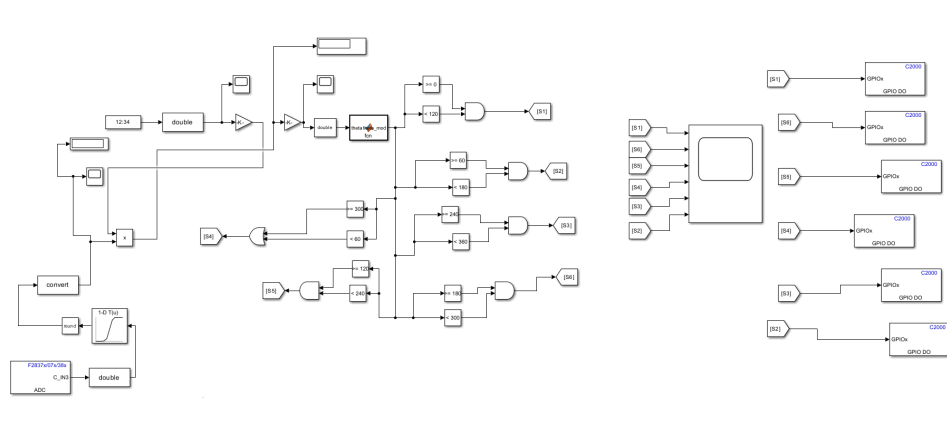


Figure 5: result

6.5 Logic-3

- To overcome the limitations observed in the previously developed control approaches, a new control logic was designed and implemented for the BLDC motor drive system. The proposed logic was developed by modifying the switching strategy and incorporating the required control methodology necessary for proper motor operation. Unlike the earlier implementations, the new approach ensured correct commutation sequence generation, appropriate gate pulse control, and improved synchronization between the inverter switching and motor operation.

The developed control logic was tested in the simulation environment, and the obtained results confirmed the successful operation of the BLDC motor. The motor was able to start smoothly and operate under the desired conditions with stable performance characteristics. The simulation waveforms verified proper switching operation, correct phase excitation, and satisfactory motor response. Thus, the newly developed logic successfully addressed the issues present in the earlier control methods and enabled effective operation of the BLDC motor drive system.



10. Project Demonstration and Response

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Ramachandra sir , for his mentorship and technical guidance throughout the project. I also extend my gratitude to the teaching assistants **Surya Kiran** sir ,**Surya Prakash Yadav** sir, **THOITHOI ANGOM** sir , Lab Assistant-Abhishek sir and special thanks to professor- **Reddy C. C.** sir (HOD, Electrical Department) for their continuous support.

Looking forward to further exploring advanced motor control strategies and high-efficiency power conversion systems.

#PowerElectronics #MotorControl #BLDC #EmbeddedSystems #InverterDesign #ControlSystems #IITRopar #EngineeringInnovation #Sedmac #Automberg #LamResearch #Texasinstrument #Eaton #Stm #RockwellAutomation #Mistubishi #Deltaelectronics #Getmysolutions #Rigol



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The developed BLDC motor drive system and its simulation results were also demonstrated through a video presentation uploaded on LinkedIn. The uploaded demonstration received a positive response from viewers and gained approximately 244 likes, which reflects the technical relevance and practical interest in the developed work. The response on the platform motivated further improvement and validation of the proposed control logic for the BLDC motor drive system.

11. Conclusion

In conclusion, different control approaches for the BLDC motor drive system were developed and analyzed during this work. Initially, the control logic was implemented using the SPWM technique for 120-degree mode operation. Later, based on further discussion and analysis, the control method was modified according to the standard six-step commutation technique used in conventional 120-degree conduction mode. However, the motor could not achieve proper operation due to the absence of an appropriate control strategy.

To resolve this issue, a new control logic was developed and implemented successfully. The proposed method provided proper switching control and enabled successful operation of the BLDC motor in the simulation environment. The obtained results confirmed stable motor operation, correct commutation sequence, and satisfactory system performance. The simulation waveforms and motor response validated the effectiveness of the developed control approach.

The corresponding simulation and motor operation results are attached in the form of a video for reference and verification.